SAGAR PUBLIC SCHOOL

Integrated Programme | Topic: Electrostatics – Miscellaneous & PYQs

PRACTICE SHEET 4

Miscellaneous and PYQs covering the entire chapter

Time Allowed: 60 Mins	**Total Questions: 20**

Name: _____ Class/Sec: _____ Date: _____

Instructions: Attempt all questions. Each question carries one correct option unless marked *one or more than one option is correct.* Questions without options are numerical-value type: write the answer in the blank. The tag after each question shows its source examination. A self-assessment page follows; the answer key is provided as a separate sheet.

Q.1 A point charge $q = 1\,\mu\text{C}$ is located at a distance 2 cm from one end of a thin insulating wire of length 10 cm having a charge $Q = 24\,\mu\text{C}$, distributed uniformly along its length, as shown in figure. Force between q and wire is _____ N. (Use: $\dfrac{1}{4\pi\epsilon_0} = 9 \times 10^9\ \text{N}\cdot\text{m}^2/\text{C}^2$) *[JEE Main 2026, 24 Jan, Eve.]*

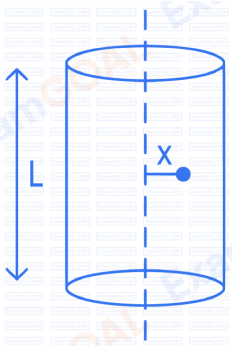


Q.2 A point charge $q_1 = 4q_0$ is placed at origin. Another point charge $q_2 = -q_0$ is placed at $x = 12$ cm. Charge of proton is q_0. The proton is placed on x axis so that the electrostatic force on the proton is zero. In this situation, the position of the proton from the origin is _____ cm. *[JEE Main 2023, 29 Jan, Mor.]*

Q.3 An infinite number of point charges, each carrying $1\,\mu\text{C}$ charge, are placed along the y-axis at $y = 1$ m, 2 m, 4 m, 8 m ... The total force on a 1 C point charge, placed at the origin, is $x \times 10^3$ N. The value of x, to the nearest integer, is _____. [Take $\dfrac{1}{4\pi\epsilon_0} = 9 \times 10^9\ \text{Nm}^2/\text{C}^2$] *[JEE Main 2021, 18 Mar, Eve.]*

Q.4 Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle of 37° with each other. When suspended in a liquid of density 0.7 g/cm^3, the angle remains same. If density of material of the sphere is 1.4 g/cm^3, the dielectric constant of the liquid is _____. $\left(\tan 37^\circ = \dfrac{3}{4}\right)$ *[JEE Main 2024, 30 Jan, Eve.]*

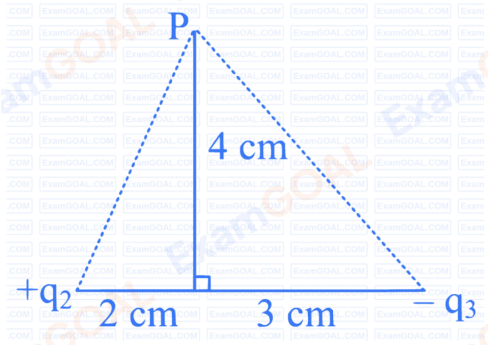
Q.5 A long cylindrical volume contains a uniformly distributed charge of density $\rho\ \text{C m}^{-3}$. The electric field inside the cylindrical volume at a distance $x = \dfrac{2\epsilon_0}{\rho}$ m from its axis is _____ V m^{-1}. *[JEE Main 2022, 27 Jul, Mor.]*



Q.6 An infinite plane sheet of charge having uniform surface charge density $+\sigma_s$ C/m^2 is placed on x-y plane. Another infinitely long line charge having uniform linear charge density $+\lambda_e$ C/m is placed at $z = 4$ m plane and parallel to y-axis. If the magnitude values $|\sigma_s| = 2|\lambda_e|$ then at point $(0, 0, 2)$, the ratio of magnitudes of electric field values due to sheet charge to that of line charge is $\pi\sqrt{n} : 1$. The value of n is _____. *[JEE Main 2024, 4 Apr, Mor.]*

Q.7 The electric field in a region is given by $\vec{E} = (2\hat{i} + 4\hat{j} + 6\hat{k}) \times 10^3$ N/C. The flux of the field through a rectangular surface parallel to x–z plane is 6.0 Nm^2C^{-1}. The area of the surface is _____ cm^2. *[JEE Main 2025, 7 Apr, Eve.]*

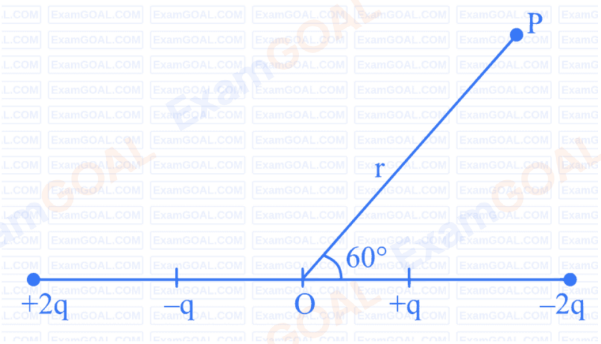
Q.8 If the net electric field at point P along Y axis is zero, then the ratio of $\left|\frac{q_2}{q_3}\right|$ is $\frac{8}{5\sqrt{x}}$, where $x =$ _____. *[JEE Main 2024, 8 Apr, Eve.]*



Q.9 64 identical drops each charged upto potential of 10 mV are combined to form a bigger drop. The potential of the bigger drop will be _____ mV. *[JEE Main 2023, 12 Apr, Mor.]*

Q.10 For a charged spherical ball, electrostatic potential inside the ball varies with r as $V = 2ar^2 + b$. Here, a and b are constant and r is the distance from the center. The volume charge density inside the ball is $-\lambda a\varepsilon$. The value of λ is _____. (ε = permittivity of the medium) *[JEE Main 2023, 29 Jan, Eve.]*

Q.11 The distance between charges $+q$ and $-q$ is $2l$ and between $+2q$ and $-2q$ is $4l$. The electrostatic potential at point P at a distance r from center O is $-\alpha\left[\frac{ql}{r^2}\right] \times 10^9$ V, where the value of α is _____. (Use $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ Nm^{2}C^{-2}) *[JEE Main 2024, 31 Jan, Eve.]*



Q.12 27 similar drops of mercury are maintained at 10 V each. All these spherical drops combine into a single big drop. The potential energy of the bigger drop is _____ times that of a smaller drop. *[JEE Main 2021, 26 Feb, Eve.]*

Q.13 An electric dipole of dipole moment 6×10^{-6} Cm is placed in uniform electric field of magnitude 10^6 V/m. Initially, the dipole moment is parallel to electric field. The work that needs to be done on the dipole to make its dipole moment opposite to the field, will be _____ J. *[JEE Main 2025, 28 Jan, Eve.]*

Q.14 Two charges of $-4\,\mu$C and $+4\,\mu$C are placed at the points A$(1, 0, 4)$ m and B$(2, -1, 5)$ m located in an electric field $\vec{E} = 0.20\,\hat{i}$ V/cm. The magnitude of the torque acting on the dipole is $8\sqrt{\alpha} \times 10^{-5}$ Nm, where $\alpha =$ _____. *[JEE Main 2024, 27 Jan, Eve.]*

Q.15 A particle of mass 1 mg and charge q is lying at the mid-point of two stationary particles kept at a distance 2 m when each is carrying same charge q. If the free charged particle is displaced from its equilibrium position through distance x ($x \ll 1$ m), the particle executes SHM. Its angular frequency of oscillation will be _____ $\times 10^5$ rad/s if $q^2 = 10$ C^2. *[JEE Main 2021, 25 Jul, Mor.]*

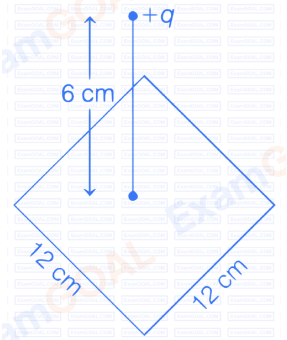
Q.16 *(One or more than one option is correct)* A uniformly charged solid sphere of radius R has potential V_0 (measured with respect to ∞) on its surface. For this sphere the equipotential surfaces with potentials $\frac{3V_0}{2}$, $\frac{5V_0}{4}$, $\frac{3V_0}{4}$ and $\frac{V_0}{4}$ have radius R_1, R_2, R_3 and R_4 respectively. Then *[JEE Main 2015 (Offline)]*

(A) $R_1 = 0$ and $R_2 < (R_4 - R_3)$
(B) $2R < R_4$
(C) $R_1 = 0$ and $R_2 > (R_4 - R_3)$
(D) $R_1 \neq 0$ and $(R_2 - R_1) > (R_4 - R_3)$

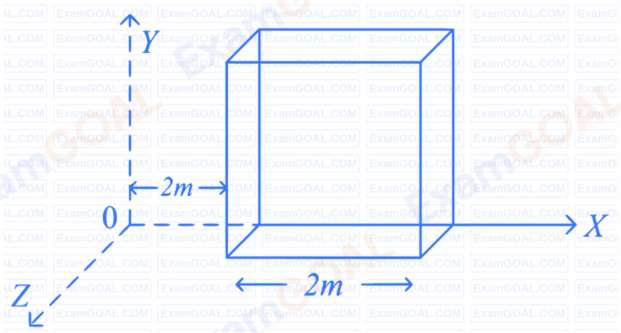
Q.17 Two identical conducting spheres with negligible volume have 2.1 nC and -0.1 nC charges, respectively. They are brought into contact and then separated by a distance of 0.5 m. The electrostatic force acting between the spheres is _____ $\times 10^{-9}$ N. [Given: $4\pi\epsilon_0 = \frac{1}{9 \times 10^9}$ SI unit] *[JEE Main 2021, 25 Feb, Eve.]*

Q.18 Three concentric spherical metallic shells X, Y and Z of radius a, b and c respectively $[a < b < c]$ have surface charge densities σ, $-\sigma$ and σ respectively. The shells X and Z are at same potential. If the radii of X & Y are 2 cm and 3 cm, respectively. The radius of shell Z is _____ cm. *[JEE Main 2023, 10 Apr, Mor.]*

Q.19 A point charge of $+12\,\mu$C is at a distance 6 cm vertically above the centre of a square of side 12 cm as shown in figure. The magnitude of the electric flux through the square will be _____ $\times 10^3$ Nm^2/C. *[JEE Main 2021, 24 Feb, Eve.]*



Q.20 An electric field $\vec{E} = (2x\,\hat{i})$ NC^{-1} exists in space. A cube of side 2 m is placed in the space as per figure given below. The electric flux through the cube is _____ Nm^2/C. *[JEE Main 2024, 9 Apr, Eve.]*



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SELF-ASSESSMENT – PRACTICE SHEET 4

Time Allowed: 60 Mins	**Time Taken:** _____ Mins
Total Questions: 20	**Questions Attempted:** _____
Correct: _____	**Incorrect:** _____
Unattempted: _____	**Accuracy:** _____ % (Correct / Attempted ×100)

Question-wise record (tick one: C = Correct, W = Wrong, S = Skipped)

Q.1 ☐C ☐W ☐S	**Q.2** ☐C ☐W ☐S	**Q.3** ☐C ☐W ☐S	**Q.4** ☐C ☐W ☐S
Q.5 ☐C ☐W ☐S	**Q.6** ☐C ☐W ☐S	**Q.7** ☐C ☐W ☐S	**Q.8** ☐C ☐W ☐S
Q.9 ☐C ☐W ☐S	**Q.10** ☐C ☐W ☐S	**Q.11** ☐C ☐W ☐S	**Q.12** ☐C ☐W ☐S
Q.13 ☐C ☐W ☐S	**Q.14** ☐C ☐W ☐S	**Q.15** ☐C ☐W ☐S	**Q.16** ☐C ☐W ☐S
Q.17 ☐C ☐W ☐S	**Q.18** ☐C ☐W ☐S	**Q.19** ☐C ☐W ☐S	**Q.20** ☐C ☐W ☐S

Reason for mistakes (tick all that apply)

☐ Concept not clear ☐ Formula forgotten ☐ Calculation mistake

☐ Misread the question ☐ Ran out of time ☐ Guess work

Personal Feedback

What went well:

_____

_____

Topics / questions I need to revise:

_____

_____

My action plan before the next sheet:

_____

_____

Teacher's remarks:

_____

_____

Student's signature: _____ Teacher's signature: _____